

The Kinetics of Choice: An Operant Systems Analysis

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The behaving organism and its environment constitute interacting elements of a feedback system. Starting with a kinetic model of the effects of reinforcement on preference and incorporating feedback functions describing the effects of responding on reinforcement rate, a complete systems analysis of concurrent reinforcement schedule situations was developed. Correct predictions were generated of both the form of the acquisition of preference and of asymptotic preference levels for choice between probabilistic reinforcement schedules, that is, variable interval and variable ratio. A kinetic model was then formulated at the level of switching behavior. All of the predictions at the molar (preference) level followed also from this molecular analysis. In addition, the switching model correctly predicted a constant sum of transition probabilities (local switching rates) and a decrease in overall switching rate as both preference strength and the difference between reinforcement rates increased. The present systems analysis is unique among recent theoretical efforts concerned with schedules of reinforcement in its ability to describe molecular as well as molar properties of both transition-state and steady-state behavior.

An operant may be functionally defined as a class of responses that is controlled by its consequences (Catania, 1968). Implicit in this definition is a view of the organism and its environment as interacting elements of a feedback system: Responses operate on the environment, and the resultant environmental changes feed back and modify the organism's behavior (Baum, 1973). This conception is not new, but only recently have theorists stressed the need to deal with feedback in a quantitative way (Baum, 1973; Heyman & Luce, 1979; Prelec & Herrnstein, 1978; Rachlin, 1978; Staddon & Motheral, 1978). As Prelec and Herrnstein (1978) pointed out, functional relationships between reinforcement rates and

response rates such as the matching law (Herrnstein, 1961) contain no true independent variables. Reinforcement rates are not usually specified by the experimenter independently of behavior. Instead the experimenter, in specifying a reinforcement schedule, specifies a rule by which behavior determines reinforcement rate. The mathematical expression of such a relationship between reinforcement rate and responding is called a *feedback function* (Baum, 1973). The use of feedback functions is central to several recent developments in mathematical behavior theory (Baum, 1973; Heyman & Luce, 1979; Prelec & Herrnstein, 1978; Rachlin, 1978; Staddon & Motheral, 1978).

In the area of animal choice behavior, feedback functions have led to new insights regarding steady-state preference (Prelec & Herrnstein, 1978), and their exact formulation is fundamental to the question of the relationship between reinforcement maximization and matching in the steady state (Heyman & Luce, 1979; Staddon & Motheral, 1978). As Baum (1973) noted, however, the analysis of feedback systems in other disciplines attempts

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to describe the transient response to changes in input as well as the equilibrium or steady-state behavior. In fact, the equilibrium behavior of a system is implicit in a successful description of the transient response (Milhorn, 1966; Rosen, 1970). Although historically the study of operant behavior has concentrated on steady-state performance, systems analysis techniques promise quantitative accounts of responding in both transition and steady states. By the term *systems analysis*, we mean not just control theory (e.g., Powers, 1978) and linear systems analysis (e.g., McDowell & Kessel, 1979), but rather the entire range of mathematical and simulation techniques for analysis of the dynamics of whole systems.

Staddon (1977) recently proposed a model of the kinetics of the reinforcement process. Although Staddon emphasized the effects of reinforcement on response strength, the kinetic model may be elaborated into a complete systems analysis by incorporating mathematical descriptions of the effects of behavior on reinforcement rates. We will demonstrate this later by considering the application of the kinetic model to concurrent reinforcement schedules (Ferster & Skinner, 1957). Studies of human choice behavior have typically used discrete trials procedures, and therefore the corresponding mathematical theory formulation has been based on difference equations. Consideration of animal choice behavior on probabilistic reinforcement schedules, however, enables us to use ordinary differential equations, the traditional tools of dynamical systems analysis (Rosen, 1970).

There is a large literature concerning animal choice behavior on concurrent schedules of reinforcement (see de Villiers, 1977, for a review). Two types of concurrent schedule procedures are used, but both methods program reinforcements separately for different response alternatives. In "two-key" procedures, subjects may switch back and forth between two operanda, each associated with a different reinforcement schedule. In "changeover-key" procedures, subjects may switch between two schedules programmed on the same operandum by responding on a second changeover operandum; each schedule is associated with a different stimulus. With both types of concurrent schedule procedures, a "changeover

delay" (COD) contingency is often used. The COD value specifies the minimum time interval that must elapse between a switch to an alternative and reinforcement for that alternative.

We will consider (a) concurrent variable-interval (VI) schedules in which the VI value for each schedule specifies the average minimum interreinforcement interval, (b) concurrent variable-ratio (VR) schedules in which the VR value for each schedule specifies the average number of responses per reinforcement, and (c) concurrent VI-VR schedules. These situations will be analyzed at the *molar* level in which the dependent variables are the average preference values for experimental sessions, and predictions of both transition-state and steady-state behavior will be developed and evaluated. Then a kinetic model will be formulated at the *molecular* level in which the dependent variables are the instantaneous rates of switching from one schedule to another. It will be shown that the predictions of molar phenomena may also be derived from the molecular or switching model. Predictions of properties of switching behavior that are not directly reflected in molar preference values will also be derived.

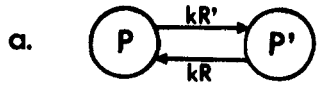
The Kinetic Model and the Matching Law

The results of a large number of experiments on concurrent schedules are well described by the matching law (Herrnstein, 1961),

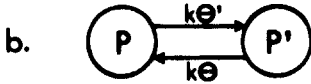
$$B_1/(B_1 + B_2) = R_1/(R_1 + R_2), \quad (1)$$

where B_n is the rate of response or amount of time allocated to some behavior n and R_n is the obtained rate of reinforcement for this behavior. In Equation 1 the relative rate of a response or the relative amount of time allocated *matches* its relative reinforcement rate. This equation is especially successful at describing steady-state choice between probabilistically reinforced alternatives, that is, VI and VR schedules (de Villiers, 1977).

Starting with the basic assumptions of Staddon's (1977) model of the kinetics of the reinforcement process, the derivation of the matching law may be reformulated. Staddon suggested that the effect of reinforcement of a response is to decrease the strengths of alter-



GENERAL KINETIC MODEL



CONC VI 1/Θ VI 1/Θ'



CONC VR 1/φ VR 1/φ'



CONC VR 1/φ VI 1/Θ

Figure 1. Kinetic models. (The conversion of preference [P] for the response reinforced at rate R to preference [P'] for the alternative reinforced at rate R' , as well as the reverse reaction, conversion of P' to P , are diagrammed. VI = variable interval; VR = variable ratio. The expression next to an arrow governs the rate of the conversion. Represented are [a] the general model stated in terms of obtained reinforcement rates, R and R' ; [b] the model of concurrent [conc] VI-VI situations assuming the form of the VI feedback function to be $R \propto \theta$; [c] the model of conc VR-VR situations assuming the form of the VR feedback function to be $R = Pr\phi$, where P is preference measured as relative time allocation and r is the local response rate; and [d] the model of conc VI-VR situations assuming the preceding feedback functions.)

native responses in proportion to their current strengths. This assumption is similar to that of earlier linear models such as those of Estes (1950b) and Bush and Mosteller (1951). We will assume a similar effect of reinforcement on preference defined as *instantaneous* relative probability; in practice, however, preference is measured as the relative frequency of re-

sponses or as relative time allocation recorded over some finite time interval. In the two-alternative case, reinforcing a response for which preference is $P_1 = B_1/(B_1 + B_2)$ decreases preference for the alternative, P_2 . The rate at which P_2 decreases will be proportional to the rate R_1 , at which the first response is reinforced. Thus,

$$dP_2/dt = -kR_1P_2, \quad (2)$$

where the constant k governs the proportion of change per reinforcement.

Because preference is defined as *relative probability*, $P_1 + P_2 = 1$, and decreasing the preference for one alternative must increase the preference for the other. Therefore, the decrease in P_2 in Equation 2 implies an equal and opposite reaction, a compensatory increase in P_1 :

$$dP_1/dt = +kR_1P_2. \quad (3)$$

If, however, in addition to reinforcing one response at rate R_1 , we also reinforce the other response at rate R_2 , then Equation 3 must be supplemented to include the proportional decrease in P_1 produced by reinforcement of the second response. Moreover, Equation 2 must be supplemented to include the compensatory increase in P_2 that comes from decreasing P_1 . Therefore, the complete system is described by

$$dP_1/dt = kR_1P_2 - kR_2P_1 \quad (4a)$$

and

$$dP_2/dt = kR_2P_1 - kR_1P_2. \quad (4b)$$

This system is represented diagrammatically in Figure 1a. At equilibrium,

$$dP_1/dt = dP_2/dt = 0,$$

whereupon Equations 4a and 4b both lead to

$$R_1/R_2 = P_1/P_2 = \frac{B_1/(B_1 + B_2)}{B_2/(B_1 + B_2)} = B_1/B_2,$$

which is algebraically equivalent to Equation 1 (Baum & Rachlin, 1969).

For purposes of solving the system (Equations 4a and 4b) as we will do for several cases considered next, it is desirable to rewrite it in terms of a single variable, P . Recalling that $P_1 + P_2 = 1$, we may obtain from Equations 4a and 4b for either response,

$$dP/dt = kR(1 - P) - kR'P, \quad (5)$$

where R is the reinforcement rate for the response whose relative frequency is P and R' the reinforcement rate for the alternative response. This formulation, Equation 5, is equivalent to Staddon's (1977) Equation 10. It should be noted that Equation 5 is not, in general, linear in P because reinforcement rate is often some function of response rate.

The general kinetic model just described is a hypothesis concerning the functional relation between the reinforcement input to the organism and the behavioral output. This kind of functional relation has been termed an "O-rule" by Baum (1973) and would be called a "transfer function" by an engineer (Milhorn, 1966). Although the O-rule is the transfer function of principal interest to a psychologist, a second transfer function, termed an "E-rule" by Baum, must be stated for a complete description of the operant feedback system. This E-rule is the feedback function that specifies how input to the organism will vary in response to changes in the organism's output.

To obtain predictions of transition-state behavior from the kinetic model, we first formulate the E-rules or feedback transfer functions governing the effect of responding on reinforcement rate. Then we substitute these feedback functions for the obtained reinforcement rates, R and R' in the above differential equation, and obtain the solutions. The solutions will be functions that describe the predicted time course of concurrent responding following step changes in either the schedule parameters or the form of the E-rules.

Concurrent VI-VI

We shall consider first the concurrent (conc) VI-VI situation that provided the original basis for the matching law (Herrnstein, 1961) and on which the bulk of experimentation with concurrent schedules has been done. Treatment of this case is facilitated by the fact that at the typical rates of switching between schedules, obtained rates of reinforcement are relatively insensitive to variations in the allocation of behavior (Heyman & Luce, 1979). This fact has special significance for a systems analysis. It means that with the conc VI-VI procedure, the effects of feedback may be minimized, and one may observe the "open-

loop" behavior of the system. As such decoupling of the feedback loop is a fundamental simplifying step in the analysis of feedback systems (Milhorn, 1966), it is not surprising that the experimental analysis of choice behavior has concentrated on this procedure.

We shall assume, therefore, that for a VI $1/\theta$ with a maximum reinforcement rate of θ , $R \simeq \theta$. Although this is an idealization, the test of whether a simplifying assumption neglects necessary descriptive detail lies in the accuracy of the predictions made possible by the assumption. In the present case, we hope to show that the behavior of the idealized system, diagrammed in Figure 1b, corresponds extremely well to the actual performance of organisms on concurrent reinforcement schedules.

Substituting θ for R in Equation 5, we obtain for either response

$$\begin{aligned} dP/dt &= k\theta(1 - P) - k\theta'P \\ &= k\theta - k(\theta + \theta')P, \end{aligned} \quad (6)$$

where θ is the reinforcement rate for the response whose relative frequency equals P , and θ' is the reinforcement rate for the alternative response. The solution to Equation 6 is

$$P(t) = Ce^{-(\theta + \theta')kt} + \theta/(\theta + \theta'), \quad (7)$$

where

$$C = P(0) - \theta/(\theta + \theta'),$$

that is, the difference between the initial value, $P(0)$, and the equilibrium value of P . As t increases, the value of $C \exp(-(\theta + \theta')kt)$ decreases and P approaches $\theta/(\theta + \theta')$, from above if $P(0) > \theta/(\theta + \theta')$ so that C is positive, and from below if $P(0) < \theta/(\theta + \theta')$ so that C is negative. Because Equation 7 is an exponential, the difference between the current and equilibrium values of P is decreased by a constant proportion in each successive unit of time.

In addition to predicting the form of the transient response, Equation 7 also predicts the effects of varying reinforcement parameters. The rate constant of the exponential decay determines the rate at which the system accomplishes the transition from initial to equilibrium values. The rate constant in Equation 7 is $k(\theta + \theta')$, indicating (as noted by Staddon, 1977) that the speed of the system's transient response is proportional to the overall rate of reinforcement. In general,

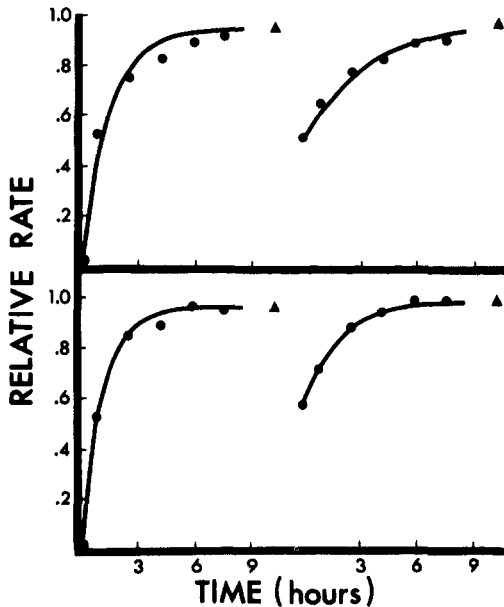


Figure 2. Development of preference. (Values were calculated from data for Catania's [1969] experiment. A two-key procedure with variable interval [VI] 180-sec and extinction schedules was used. For the top two graphs, the changeover delay [COD] was 0.5 sec; for the two bottom graphs, the COD value was 1 sec. For the two graphs on the left, the keys associated with the VI and extinction [ext] components were reversed in the preceding condition. For the two graphs on the right, the preceding condition was a conc VI 180-sec VI-180-sec. The theoretical curve is Equation 7 with k , the only free parameter, equal to the value that produced the least squares best fit.)

the time required to accomplish a given proportion of the transition to equilibrium will depend on overall reinforcement rate and not on either the distribution of reinforcements between alternative schedules or the difference between initial and equilibrium values.

These predictions were tested against the transition-state data available in the literature. Catania (1969) observed the effect of reinforcing only one response following reinforcement of both alternatives in the previous condition of the experiment. He also reported the effect of reinforcing one response after reinforcing only the alternative in the previous condition. Catania used VI 3-min schedules in a two-key procedure with either a .5-sec or a 1-sec COD. Each published data point represented the mean for two birds for a 100-min session. The data were fit using Equation 7

with k as the only free parameter. $P(0)$ was set equal to the preference observed in the session prior to the change in conditions, and the asymptote was fixed at the value observed in the 14th and final session. As may be seen in Figure 2, Catania's results are consistent with the kinetic model's prediction of an exponential decrease in preference for the unreinforced alternative.

The exponential form of the transient response of the system is not peculiar to the development of extreme preference nor is it the consequence of averaging data from different subjects. The kinetic model's predictions are also consistent with individual subject data reported by Killeen (1972). Killeen employed a changeover-key procedure with a 2-sec COD and VI 40-sec and VI 120-sec schedules. Two sessions whose average duration was between 17 and 18 min were run daily. The colors of the keys associated with the different schedules were reversed twice. The form of the transient response shown in Figure 3 is well described

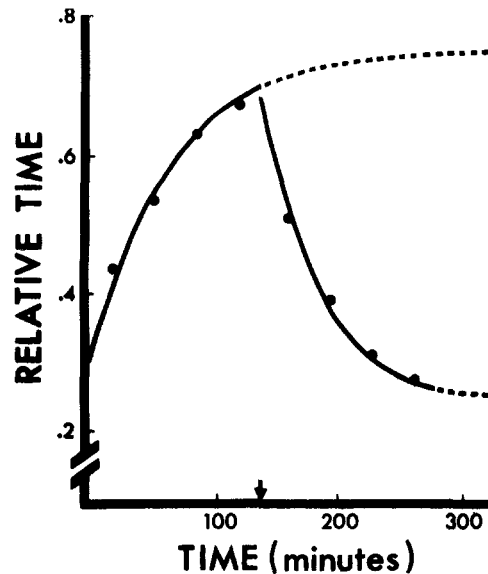


Figure 3. Development of preference. (Values were calculated from data for Bird 31 in Killeen's, 1972, experiment. A changeover key procedure with variable interval [VI] 40-sec and VI 120-sec schedules was used. The key colors associated with the different schedules were reversed twice, once at Time 0 and once at the point indicated by the arrow. The theoretical curve is Equation 7 with k , the only free parameter, equal to the value that produced the least-squares best fit.)

by Equation 7 with k as the only free parameter: $P(0)$ was set equal to the preference, measured as relative time spent in a given key color, observed in the session prior to the change in conditions, and preference was assumed to approach an asymptote at $\theta/(\theta + \theta')$.

The results of the Catania (1969) and Killeen (1972) experiments suggest that the kinetic model describes the acquisition of preference on conc VI-VI schedules regardless of differences in type of procedure, measure of preference (time or response based), COD values, session duration, initial and asymptotic preference levels, and both overall rate and distribution of reinforcements. These studies also support the prediction that the rate at which preference is acquired depends on the overall rate of reinforcement. The obtained reinforcement rates in Killeen's experiment were about five times that in Catania's experiment, and preference was acquired nearly four times as fast as in the latter study; acquisition for Bird 36 in Killeen's experiment was not shown in Figure 3 because it occurred so rapidly as to conceal the form of the transient response. The similarity of k values for the two studies reflects the fact that the difference in speed of acquisition was primarily due to the difference in reinforcement rates. The mean value of k was .030 for Catania's experiment and .022 for both birds in Killeen's experiment with a standard deviation of .011 in both cases. This degree of correspondence is remarkable, considering the substantial differences in experimental procedures, and indicates similar proportions of behavior change per reinforcement in the two studies.

Conc VR-VR

Unlike conc VI-VI schedules, which result in exclusive preference for one response only when the alternative is never reinforced, conc VR-VR schedules almost always result in exclusive preference unless the VR values are equal. Exclusive preference might be considered a trivial instance of the matching law. From the point of view of the kinetic model, however, all matching is the necessary consequence of the effect of reinforcement on preference, and the differences between the conc VI-VI and

conc VR-VR cases must be attributable to the differences in the effect of responding on rate of reinforcement (Figures 1b and 1c).

The obtained reinforcement rate on VR schedules is proportional to the response rate, and the feedback function for a VR schedule in a concurrent situation is

$$R = r\phi P,$$

where ϕ is the ratio of reinforcements to responses specified by the schedule, r is the local response rate, and P is preference measured as the proportion of time allocated to the VR schedule. We may now proceed as we did in the conc VI-VI case; substituting $r\phi P$ for R in Equation 5, we obtain for either response:

$$\begin{aligned} dP/dt &= kr\phi P(1 - P) - kr\phi'(1 - P)P \\ &= kr(\phi - \phi')P(1 - P), \quad (8) \end{aligned}$$

where ϕ and ϕ' are the reinforcement to response ratios for the response of preference P and its alternative, respectively. The solution to this equation is the logistic function

$$P(t) = P(0) \frac{1 + C}{1 + Ce^{-kr(\phi - \phi')t}}$$

where

$$C = \frac{1 - P(0)}{P(0)},$$

which is the ratio of the initial preference values for the two alternatives. The difference between the two schedule parameters, ϕ and ϕ' , determines both the rate and the direction of change in P . If $\phi > \phi'$, the expression $C \exp(-kr(\phi - \phi')t)$ is a decreasing function of time, and as it approaches zero, $P(t)$ approaches 1; if $\phi < \phi'$, the exponential expression is an increasing function of time, and as it approaches infinity, $P(t)$ approaches zero; if $\phi = \phi'$, the exponential expression equals C for all values of t , implying that preference will not change from its initial value.

The sigmoidal shape of the logistic function results from the interaction of positive feedback with the damping effect of the constraint that $P_1 + P_2 = 1$. If the rate of reinforcement for one response of preference P_1 exceeds the reinforcement rate for the alternative, P_1 increases and P_2 decreases. More time is then allocated to the first response further exaggerating the difference in reinforcement rates. At the same time, however, the size of the in-

crement in P_1 per reinforcement keeps decreasing because such increments are proportional to the dwindling size of P_2 .

Estes (1950a) studied the behavior of rats in a two-lever experimental chamber when continuous reinforcement for one operant was discontinued and continuous reinforcement for the second operant initiated. He found that the rate of responding for the second operant was described by a logistic function. If we assume that the sum of the rates of both operants remains constant and behavior is simply reallocated, then the acquisition of preference is also described by a logistic function. Continuous reinforcement schedules are covered by our model for choice between probabilistically reinforced alternatives as the limiting case when the probability of reinforcement equals one. Extinction (ext) marks the other limit with a reinforcement probability of zero. The Estes (1950a) study may therefore be thought of as shifting subjects from conc VR 1 ext to conc ext VR 1, and the logistic form of acquisition of the new preference supports our preceding analysis.

Although no other transition-state data have been published for the conc VR-VR situation, the kinetic model does correctly predict the steady-state results of Herrnstein and Loveland (1975). As we have seen above, when $\phi > \phi'$, P asymptotes at 1, and when $\phi < \phi'$, P asymptotes at 0; thus exclusive preference always develops for the schedule with the more favorable ratio of reinforcements to responses. When $\phi = \phi'$, the system differs from other concurrent situations in that it does not resist perturbation but will be stable at any preference value. Should there be any intrinsic variability, therefore, preference may wander unpredictably. Consistent with the preceding analysis, Herrnstein and Loveland reported that exclusive preference developed unless the VR schedules were equal or nearly so; in the latter case preferences were unpredictable but usually not exclusive, and first and second determinations often differed widely.

Conc VI-VR

Finally we will consider the conc VI-VR situation (Figure 1d). Substituting the feed-

back functions for VI and VR schedules into Equations 4a and 4b, we obtain

$$dP_1/dt = kr\phi P_1 P_2 - k\theta P_1 \quad (9a)$$

and

$$dP_2/dt = k\theta P_1 - kr\phi P_1 P_2, \quad (9b)$$

where P_1 and P_2 are the VR and VI preferences and ϕ and θ are the VR and VI schedule parameters, respectively. Focusing on preference for the VR schedule, substituting $1 - P_1$ for P_2 in Equation 9a and rearranging, we obtain

$$dP_1/dt = k(r\phi - \theta)P_1 - kr\phi P_1^2. \quad (10)$$

Like the differential equation for the conc VR-VR case (Equation 8), Equation 10 is a Bernoulli equation (Brand, 1966), and when $r\phi \neq \theta$, the solution for Equation 10 is

$$P_1(t) = \frac{P_1(0)Ce^{-kCt}}{C + B(1 - e^{-kCt})} \quad (11)$$

where

$$B = r\phi P_1(0),$$

the initial reinforcement rate for the VR schedule, and

$$C = \theta - r\phi,$$

the difference between the reinforcement rates for the VI and VR schedules if each were chosen exclusively. When $\theta < r\phi$, C is negative and the values of the exponential expressions in Equation 11 increase with time. As the exponentials become very large relative to C and B , $P(t)$ approaches the limit:

$$\lim_{t \rightarrow \infty} \frac{P_1(0)Ce^{-kCt}}{C + B(1 - e^{-kCt})} = \frac{P_1(0)Ce^{-kCt}}{-Be^{-kCt}} = \frac{P_1(0)C}{-B} \text{ for } C < 0.$$

Substituting $r\phi P_1(0)$ for B and $\theta - r\phi$ for C , cancelling and rearranging, we may obtain:

$$\lim_{t \rightarrow \infty} P_1(t) = 1 - \frac{\theta}{r\phi}$$

for the preference for the VR schedule at equilibrium. On the other hand, when $\theta > r\phi$, then C is positive and the values of the exponential expressions in Equation 11 decrease with time. As the exponentials become very small, the numerator in Equation 11 approaches zero and exclusive preference for the VI

schedule develops. Exclusive preference for the VI also develops when $r\phi = \theta$; in this case, the solution to Equation 10 is

$$P_1(t) = \frac{P_1(0)}{1 + Bkt},$$

which approaches zero as t becomes very large.

In the conc VI-VR situation, therefore, when the reinforcement rate for responding only on the VI schedule equals or exceeds the reinforcement rate for responding only on the VR schedule, exclusive preference should develop. This result has been observed by Herrnstein and Heyman (1979) using a two-key procedure. In this study it may be seen most clearly for Bird 3 who showed the least variability in local VR response rate ($M = 116.2$ responses/min, $SD = 8.23$) between conditions. This subject showed exclusive preference for the VI component of a conc VI 15-sec VR 30 schedule. The obtained reinforcement rate when the subject responded only on the VI was 3.79 reinforcements/min, whereas the predicted reinforcement rate for responding only on the VR was $3.87/\text{min} \pm .17$. Thus, when the reinforcement rates for responding only on one schedule were nearly equal, exclusive preference for the VI developed as predicted.

Herrnstein and Heyman (1979) also found that when the reinforcement rate on the VI schedule was less than that which would be obtained by responding only on the VR schedule, matching with nonexclusive preference occurs as predicted. Nonexclusive matching on conc VI-VR schedules has also been observed with a changeover-key procedure (Herrnstein & Loveland, unpublished, cited by de Villiers, 1977). In both experiments, however, preference was biased in favor of one type of schedule; birds tended to allocate more time to the VI schedule and more responses to the VR schedule than would be predicted by unbiased matching of relative reinforcement rates (Herrnstein & Heyman, 1979; Herrnstein & Loveland, unpublished, cited in de Villiers, 1977). The biased distribution of responses reflects the fact that local VR response rates were consistently higher than local VI response rates. This difference in local rates may also account for the bias in favor of the VI schedule revealed

by time allocation. Examined point by point, Herrnstein and Heyman's data show that as the ratio of VR to VI local response rates increases, time allocation becomes increasingly biased in favor of the VI schedule.

Prelec and Herrnstein (1978) described a "paradigmatic experiment" that clearly demonstrates the value of viewing concurrent schedule situations as feedback systems. Prelec and Herrnstein shifted pigeons between conc VI-VR and conc VI-VI-VR conditions by altering the availability of one of the VI keys. Staddon (1977) showed that a three-alternative situation may be described by a system of three differential equations that may be solved for P_n at equilibrium, giving

$$P_n = R_n / (R_1 + R_2 + R_3).$$

Substituting VI and VR feedback functions and focusing on the preference for the VR alternative, P_1 , we may obtain

$$P_1 = r\phi P_1 / (r\phi P_1 + \theta_a + \theta_b),$$

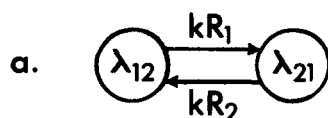
where ϕ is the ratio of reinforcements to response on the VR schedule and θ_a and θ_b are the reinforcement rates on the two VI schedules. Dividing both sides by P_1 and rearranging gives

$$P_1 = 1 - \frac{\theta_a + \theta_b}{r\phi}$$

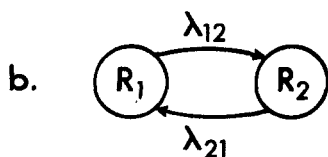
for the preference for the VR at equilibrium when $r\phi \geq \theta_a + \theta_b$. Clearly, if the value of θ_b is shifted between $\theta_b = \theta_a$ and $\theta_b = 0$, P_1 will alternately increase and decrease in value. And this is exactly what Prelec and Herrnstein observed; removing one VI from a conc VI-VI-VR with equal VIs produced a large increase in the rate of VR responding, and reintroducing the same VI schedule decreased the VR rate to its former level. As Prelec and Herrnstein pointed out, this result is of considerable theoretical significance because although it violates the "constant-ratio rule" of formal choice theory (Luce, 1959), it is correctly predicted by viewing matching as the product of a feedback system.

Molecular Analysis and the Switching Model

Pliskoff (1971) has suggested that matching, whether of time or response rate, may be a



KINETIC MODEL



STATE GRAPH

Figure 4. Kinetic model and state graph of switching behavior. (The kinetic model [a] diagrams the effect of reinforcement on local switching rates or transition probabilities. As indicated in the state graph [b], λ_{12} represents the rate of switching from an alternative reinforced at rate R_1 to a second alternative reinforced at rate R_2 , and λ_{21} represents the rate of switching back to the first alternative.)

consequence of a more fundamental process revealed in the rates of switching between alternatives. This view is supported by reports of matching when organisms choose between response-independent reinforcement schedules (Baum & Rachlin, 1969; Brownstein & Pliskoff, 1968), a situation in which switching is the only required response. The matching relation is easily restated in terms of switching behavior. If an organism switches from one alternative to a second alternative with a mean instantaneous transition probability of λ , that is, at a mean local rate of λ , then the mean duration for the first alternative equals $1/\lambda$. Therefore, the matching law (Equation 1) may be rewritten as

$$R_1/(R_1 + R_2) = (1/\lambda_{12})/(1/\lambda_{12} + 1/\lambda_{21}), \quad (12)$$

where λ_{12} is the probability of switching from the alternative reinforced at rate R_1 to a second alternative reinforced at rate R_2 , and λ_{21} is the probability of switching back to the first alternative.

We now propose a kinetic model formulated in terms of transition probabilities from which the molar kinetic model developed above may be derived. If Pliskoff's (1971) suggestion is correct, this switching model is the more fundamental of the two models. The switching model states that the effect of reinforcing an

alternative is to increase the probability of switching to that alternative and to decrease the probability of switching to the other alternative (Figure 4). Thus,

$$d\lambda_{21}/dt = kR_1\lambda_{12} - kR_2\lambda_{21} \quad (13a)$$

and

$$d\lambda_{12}/dt = kR_2\lambda_{21} - kR_1\lambda_{12}. \quad (13b)$$

At equilibrium both Equations 13a and 13b lead to

$$R_1/R_2 = (1/\lambda_{12})/(1/\lambda_{21}),$$

which is equivalent to Equation 12.

It is implicit in the idea that any increase in λ_{12} is accompanied by an equivalent decrease in λ_{21} and vice versa, as formalized in Equations 13a and 13b, that the sum of the switching probabilities always remains constant; that is, $\lambda_{12} + \lambda_{21} = c$, and the effect of reinforcement is to reallocate some portion of that fixed quantity. We may therefore restate the switching model so that for either alternative,

$$d\lambda/dt = kR'(c - \lambda) - kR\lambda, \quad (14)$$

where λ is the probability of switching from the alternative reinforced at rate R to the alternative reinforced at rate R' , and $c - \lambda$ is the probability of switching back again.

Equations 13a and 13b and Equation 14 may be seen to be isomorphic with the general kinetic model as developed at the molar level. Substitution of feedback functions for obtained reinforcement rates will, as in the molar analysis above, yield predictions of the system's transient response under different concurrent schedule conditions. The isomorphism implies that transition-state behavior as measured in terms of switching behavior will follow a trajectory of the same form as that previously predicted for preference, given the appropriate substitutions. Recalling that preference may be measured as relative time allocation, we may write

$$P(t) = \frac{1/\lambda(t)}{1/\lambda(t) + 1/(c - \lambda(t))} = 1 - \lambda(t)/c$$

and, therefore,

$$\lambda(t) = c(1 - P(t)), \quad (15)$$

where λ is the probability of switching from the alternative for which preference equals P to the alternative for which preference equals

$1 - P$. Thus the probability of switching to an alternative is at all times proportional to the preference for that alternative, and the constant of proportionality equals the sum of the switching probabilities.

We began by developing a systems analysis based on a kinetic model at the molar level. However, the preceding molecular analysis demonstrates that if we assume the effect of reinforcement on switching probabilities to be the fundamental process, then both the asymptotic preference values and the form of acquisition of preference under various concurrent schedules follow from this assumption. Moreover, the switching model generates unique predictions that may be tested against data available in the literature. One prediction unique to the switching model is that the sum of the transition probabilities is invariant across different relative reinforcement rates and preference values. This constancy has recently been reported by Heyman (1979) in a study of choice on conc VI-VI schedules using a changeover (CO)-key procedure and no COD.

We have examined the sum of transition probabilities in an experiment reported by Baum and Rachlin (1969). In this study, pigeons were reinforced according to conc VI-VI schedules for standing on one side or the other of the experimental chamber, and preference was measured as time allocation. The duration of the COD was explicitly signaled, and time accumulated during CODs was recorded separately from time spent on either side. The transition probabilities, λ_{12} and λ_{21} , each were calculated as the number of post-COD switches made from a side divided by the time allocated to that side, and the number of post-COD switches from both sides was estimated by dividing the accumulated COD time by the COD value. The results are shown in Figure 5 plotted as a function of preference strength (equal to $|P - .5| + .5$), a measure of the degree to which preference departs from indifference. It may be seen that the sum of the post-COD transition probabilities remained approximately constant over a wide range of preference strengths. Taken together with Heyman's (1979) findings, these results suggest that the switching model's

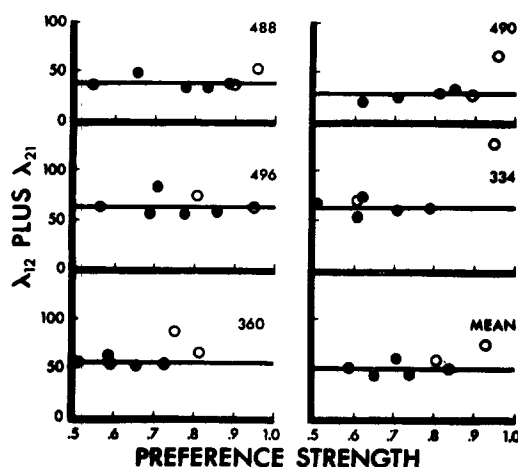


Figure 5. The sum of the transition probabilities in switches per minute as a function of preference strength. (Values were calculated from data for five subjects in Baum and Rachlin's, 1969, experiment. VI = variable interval. Preference strength measures the extent to which preference departs from indifference [.5]. Solid circles indicate the mean of multiple determinations with a single concurrent [conc] VI-VI schedule; open circles indicate the mean for single determinations with two conc VI-VI schedules for which the ratios of programmed reinforcement rates were equal, e.g., conc VI 1-min VI-8-min and conc VI .5-min VI-4-min. Horizontal lines represent the overall means of the values indicated by solid circles. These overall means provide estimates of the value of c and were used in normalizing the data presented in Figure 6.)

constancy prediction may be valid for diverse operants and experimental procedures.

Heyman (1979) has suggested that a constant sum of switching probabilities will determine the relationship between overall switching rate and preference. He considered the special case where the switching probabilities are stationary and sum to unity, but his suggestion may be shown to be true in general. The average duration of a switching cycle is $1/\lambda + 1/(c - \lambda)$. Each cycle involves a switch to the first alternative and a switch from the first alternative to the second alternative, that is, two switches per cycle. Thus, the overall switching rate, S , is given by

$$S(\lambda) = \frac{2}{1/\lambda + 1/(c - \lambda)} = 2\lambda(c - \lambda)/c,$$

which is an inverted U-shaped function with a maximum at $\lambda = c/2$, that is, at indifference. This prediction of the switching model is in

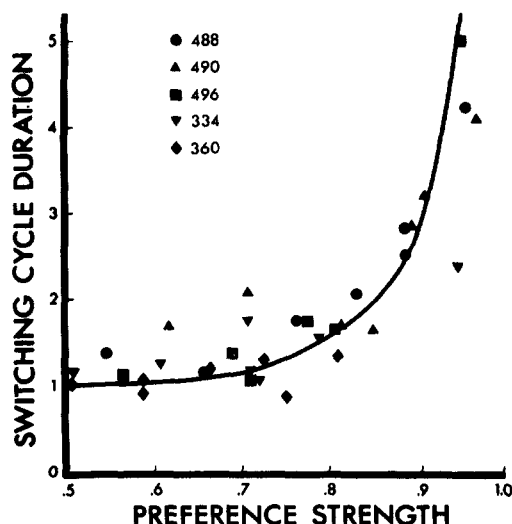


Figure 6. Normalized switching cycle duration as a function of preference strength. (Values were calculated from data for five subjects in Baum and Rachlin's, 1969, experiment. The theoretical curve is Equation 16 in the text.)

accord with numerous reports that overall switching rate on conc VI-VI schedules decreases as both preference strength and the difference between reinforcement rates increase (e.g., Herrnstein, 1961; Heyman, 1979; Stubbs & Pliskoff, 1969).

The relationship between preference strength and overall switching rate may also be stated in terms of switching cycle duration, $D = 2/S$. It is useful to normalize switching cycle duration with respect to the minimum duration observed at indifference. This minimum occurs at $\lambda = c - \lambda = c/2$ and $D_{\min} = 4/c$. Therefore, the normalized switching cycle duration is given by

$$\begin{aligned} D(\lambda)/D_{\min} &= (1/\lambda + 1/(c - \lambda))/(4/c) \\ &= c^2/4\lambda(c - \lambda). \end{aligned}$$

Recalling that $\lambda = c(1 - P)$, Equation 15, we may obtain by substitution,

$$D(P)/D_{\min} = 1/4P(1 - P). \quad (16)$$

In Equation 16, normalization has resulted in a parameter-free prediction of the relationship between preference and switching cycle duration. $D_{\min}$ is an empirical constant that may be estimated from the sum of the switching rates as shown in Figure 5. As may be seen in Figure 6, Equation 16 provides a good description of

switching cycles (less COD duration) in the Baum and Rachlin (1969) study.

The variables λ_{12} and λ_{21} in the switching model might be thought of as measuring the "attractiveness" of the alternatives because they govern the rates of leaving one alternative for another. The constant sum of $\lambda_{12} + \lambda_{21}$ suggests a constant amount of attraction to the various alternatives in a situation and necessarily implies that any increase in attraction to an alternative is accompanied by an increased "reluctance" to leave that alternative. It would seem reasonable that the amount (c) of attraction in a situation would be related to the overall reinforcement rate. The fact that in Figure 5 the sum of $\lambda_{12} + \lambda_{21}$ at extreme preference values tends to exceed the mean is consistent with this view: In the Baum and Rachlin (1969) experiment, the schedules that generated the most extreme preference also produced the highest overall reinforcement rates. The high values for $\lambda_{12} + \lambda_{21}$ are also reflected in the lower-than-predicted cycle durations at extreme preference values (Figure 6).

If future experiments confirm that the value of c is an increasing function of overall reinforcement rate, this result would raise the possibility that c , like Killeen's (1975; Killeen, Hanson, & Osborne, 1978) parameter A , is related to arousal. The observation that pigeons' switching rates decrease with increasing body weight is consistent with this view (Kulli, unpublished, cited in de Villiers, 1977). Reinforcement on concurrent schedules might be thought of as having two important effects: (a) contributing to the general level of arousal of the organism as described by Killeen and (b) partitioning that arousal between response alternatives as described by the kinetic model.

It should be noted that the predictions of overall switching rates and switching cycles are not restricted to the steady state. During a shift from greater preference for one alternative to greater preference for the other alternative, for example, the switching cycle duration should become first shorter and then longer, being minimal as preference passes through indifference. The sum of the transition probabilities, however, should remain constant

during the acquisition of a new asymptotic preference level.

Discussion

The application of systems analysis to concurrent schedule situations seems to us to be a logical extension of ideas implicit in the experimental analysis of behavior. Certain aspects of our approach may be unfamiliar, however, and it is important to distinguish between the means and the ends of this analysis. The theoretical content of the present treatment is embodied in the kinetic models of the effects of reinforcement on preference and switching. Systems analysis provides the mathematical techniques that enable us to develop the implications of the model and apply it to specific situations. The particular VI feedback function we have used in this analysis is an idealization (cf. Heyman & Luce, 1979) whose simplicity greatly expedites the mathematical development. Moreover, within the range in which the VI feedback function is a good approximation, the effects of reinforcement on preference are revealed relatively free of feedback effects. Although the kinetic models may be applied outside this range through the use of computer simulation, there are obvious advantages to the study of experimental situations for which the mathematical derivation of predictions is relatively straightforward.

It has been suggested that matching is the consequence of a process of maximizing reinforcement, either at the molecular (Shimp, 1966, 1969) or molar (Rachlin, Green, Kagel, & Battalio, 1976) level. This suggestion is not only intuitively appealing, but would also strengthen ties with economics and biology, which likewise assume that behavior is the consequence of optimization processes (e.g., Cody, 1974; Lea, 1978; Rapport & Turner, 1977). Recently, however, it has been demonstrated that matching on conc VI-VR schedules is inconsistent with maximization (Herrnstein & Heyman, 1979; Staddon & Motheral, 1978). Although there is some disagreement as to whether matching on conc VI-VI schedules maximizes reinforcement (Heyman & Luce, 1979; Rachlin et al., 1976; Staddon & Motheral, 1978), it is clear that as overall switching rates

increase (Heyman & Luce, 1979) or COD values decrease (Rachlin et al., 1976), the differential allocation of behavior has less and less effect on overall reinforcement rate. At typical switching rates and COD values, reinforcement rate is sufficiently insensitive to behavior allocation to have permitted us to model choice on conc VI-VI schedules while assuming complete independence.

As Staddon and Motheral (1978) point out, however, reinforcement maximization under experimental conditions may occur only when the situation is sufficiently similar to the natural one. Evolution may be assumed to select for optimal foraging strategies (see Pyke, Pulliam, & Charnov, 1977, for a review), but such strategies are optimal only in a specific context, and that context is, of course, the natural environment. From this point of view, the results of experiments on concurrent schedules suggest that organisms are adapted for foraging in environments similar to the conc VR-VR situation because steady-state performance on these schedules indisputably maximizes reinforcement rate. If one assumes that (a) the benefits obtained foraging in a given patch are proportional to the amount of foraging done in the patch, (b) foraging pays off randomly, and (c) patch depletion is not a factor, then the situation is analogous to conc VR-VR schedules. This similarity to the natural environment has been noted by biologists (Krebs, Kacelnik, & Taylor, 1978).

Our systems analysis adds a new dimension to optimization on conc VR-VR schedules by considering the acquisition of preference in such situations. The kinetic models predict that not only will organisms maximize reinforcement by developing exclusive preference for the more favorable alternative but also that the rate at which they adapt to a change in which alternative is more favorable will depend on the difference between the potential reinforcement rates offered by the alternatives. Thus, the speed with which organisms will modify their behavior in this situation is proportional to the advantage to be gained by the modification. Certainly any theory that deals with optimal foraging in changing environments must consider not only the adaptiveness of steady-state foraging patterns, but also the adaptiveness of the process by which

foraging patterns are changed. We believe that the kinetic models may prove useful in this regard.

Powers (1973, 1978) has contended that systems analysis represents a theoretical alternative to behaviorism. However, the present treatment demonstrates the compatibility of the two approaches and suggests that the techniques of systems analysis have much to offer the mathematical behavior theorist. Powers' analysis differs from ours in two important ways. First, his is based on a control theory model. Control theory is a branch of systems analysis that assumes the existence of control variables or reference values with which inputs are compared (Milhorn, 1966). Although such hypothetical reference values have their uses, they are by no means a necessity for the analysis of feedback systems (Milhorn, 1966; Rosen, 1970). Second, in opting for a "quasi-static" analysis (Powers, 1978), we believe that Powers has forsaken perhaps the most important attribute of systems analysis, its ability to describe both transition-state and equilibrium behavior. Diverse processes may result in the same equilibrium but yield different predictions of transition-state behavior (Milhorn, 1966; Rosen, 1970). Therefore, the ability to account for a system's transient response to changes in input constitutes a most important test of any model.

The roots of the present development lie in the linear models of Estes (1950b) and Bush and Mosteller (1951). The fundamental shared assumption is that reinforcement strengthens responses or associations in proportion to the difference between current and asymptotic strength. Any success of the present kinetic models is therefore a tribute to the insight of these theorists.

With respect to more recent sources, the present development at the molar level may be viewed as a synthesis of elements from the work of Staddon (1977) and Prelec and Herrnstein (1978) with the methodology of systems analysis. Staddon proposed a kinetic model of the reinforcement process from which the matching law may be derived, and Prelec and Herrnstein demonstrated that feedback functions may be substituted for obtained reinforcement rates in the matching law to predict steady-state preference. We have shown that

by substituting feedback functions into the kinetic model, one may obtain predictions of transition-state behavior as well as steady-state performance on concurrent schedules.

To summarize, this approach correctly predicts (a) that for conc VR-VR schedules, exclusive preference for the richer VR will develop unless the schedules are equal, in which case preference is unpredictable (Herrnstein & Loveland, 1975), and (b) that for conc VI-VR schedules, exclusive preference for the VI schedule will develop if the rate of reinforcement for responding only on the VR is less than that for the VI (Herrnstein & Heyman, 1979).

Furthermore, the available data on transition-state behavior are also consistent with the kinetic model. When reinforcement rate is proportional to response rate, preference acquisition is described by the logistic function (Estes, 1950a). When reinforcement rate is relatively independent of response rate, preference shows a negative exponential trajectory that asymptotes at matching of relative reinforcement rate, and the speed with which a new preference is acquired depends on the overall reinforcement rate (Catania, 1969; Killeen, 1972) as predicted by the kinetic model.

At the molecular level we have followed up suggestions by Pliskoff (1971) and Heyman (1979). Pliskoff suggested that preference reflects more fundamental patterns of switching between alternatives, and Heyman, noting that the sum of transition probabilities was approximately constant in his study of switching on conc VI-VI schedules, suggested that this constancy determines the relationship between preference strength and overall switching rate. We have formulated a kinetic model in terms of switching behavior and demonstrated that the predictions of both transition and steady-state preference developed at the molar level may be derived from this molecular analysis. Moreover, the kinetic model of switching behavior predicts the constant sum of transition probabilities noted by Heyman as well as the resultant decrease in overall switching rate as preference departs from indifference. A parameter-free prediction of the relationship observed between preference strength and switching-cycle duration (Baum & Rachlin, 1969) also follows from the switching model.

Whereas a variety of theories may all predict the same equilibrium behavior, that is, matching, our systems analysis based on the kinetic models is unique among recent theoretical developments concerned with schedules of reinforcement in its ability to account for transition-state behavior. The decrease in publication of cumulative records in recent years has been correlated with an increasing emphasis on the molar properties of steady-state performance (Poling, 1979; Skinner, 1976). We hope that our analysis of choice behavior on concurrent reinforcement schedules will not only stimulate further research and evaluation of the kinetic models but will also help rekindle a general interest in the dynamic and molecular aspects of operant behavior.

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